

Neural Speech Decoding for Brain Computer Interfaces

Advanced Project

Team leader: 문경은 | Team members: 구민주 김소영 노현이 윤시원 이재용 조연우 채보경

Abstract

This project investigates neural signatures associated with inner speech and examines **whether phonemic information can be decoded directly from EEG recordings**. Using a publicly available speech-related EEG dataset and computational resources, we developed a customized processing pipeline that combines task-specific signal preparation, visual analyses of spatiotemporal EEG dynamics, and a machine learning based phoneme classification framework optimized for decoding performance.

Through this work, we sought to assess how discriminative phonetic information is represented in non-invasive electrophysiological signals and to establish **a practical foundation for EEG-driven speech decoding brain computer interfaces (BCIs)** capable of translating covert speech-related activity into linguistic information.

Introduction

Why EEG-based Speech Decoding?



Why Start with Phonemes?



Smallest Unit of Speech: Serves as the foundational linguistic building block of spoken language.

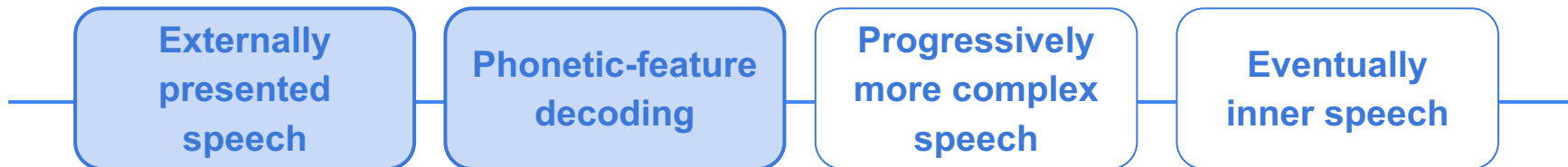
Introduction

Challenge of Inner Speech Decoding

- Unclear individual phoneme onset → difficult to determine from EEG
- Challenging alignment between neural signals and phoneme labels

A Intermediate Step

- Decoding models can first target specific properties of the speech stream
- **Externally presented speech** → timing & phoneme labels



An open-access EEG dataset for speech decoding: Exploring the role of articulation and coarticulation

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OpenNEURO



EEG dataset for speech decoding

Introduction

Goal of this project

1. To characterize how discriminative phonetic information is represented in non-invasive EEG signals.
1. To establish a preliminary basis for future EEG-based inner speech decoding by validating phonetic decoding in controlled speech tasks

Table 2.

Highest recognition rates for each subject and condition

Auditory comprehension										Internal speech													
Subject		E/O				O/E				E/O		O/E											
Best filter (Hz)			Best EEG sensors			Best filter (Hz)			Best EEG sensors			Best filter (Hz)		Best EEG sensors			Best filter (Hz)			Best EEG sensors			
Low freq		High freq	%	Loc		Low freq		High freq	%	Loc		Low freq		High freq	%	Loc		Low freq		High freq	%	Loc	
S1	2	9	51	T3	3	11	51	Cz	1.5	8.5	86	T6	1	10	91	T6							
S2	3	13	66	T4	4	9	57	T4	5	16	51	T4	5	16	54	T6							
S3	3	11	97	T3	3	11	77	T3 C3	17	23	37	FP2	5	15	37	F4							
S4	1.5	6.5	63	C3	1.5	6.5	49	C3 Cz	1.5	7	43	T6	2	8	34	T6							
S5	0.5	40	43	F4	6	12	37	C3	3	11	46	C4	2	14	46	T6							
S6	3	10	71	C4A	4	12	69	C4A															
S7	9	15	60	CZA	5	55	69	C1															

(Suppes et al., 1997; Moreira et al., 2025)

Externally
presented
speech

Phonetic-feature
decoding

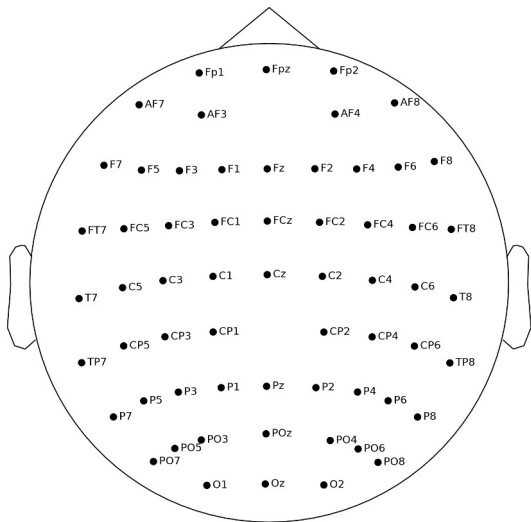
Progressively
more complex
speech

Eventually
inner speech

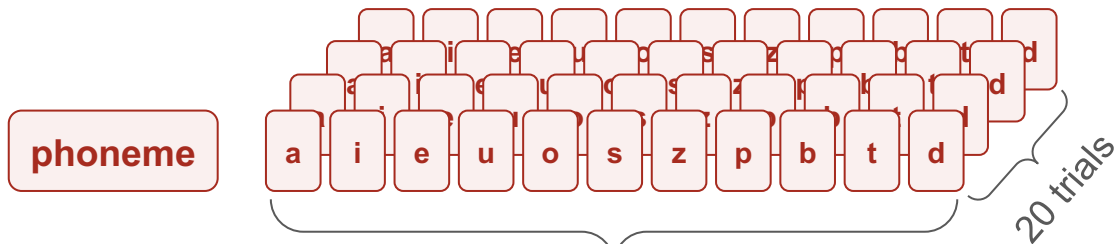
Dataset

Dataset

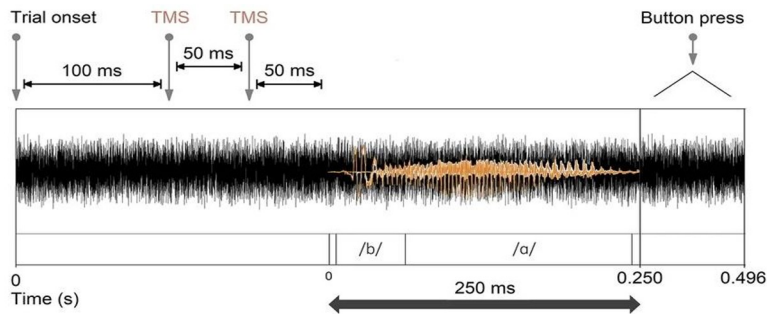
Baseline Channel Map (61 Channels)



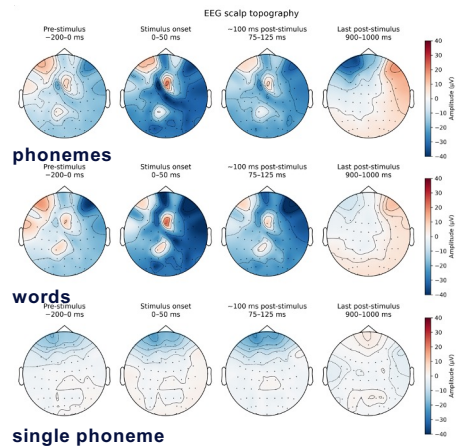
(left) 참가자 16명 | 61 channels



11개 음소
(top)



(bottom) 2,000 Hz



phonemes

words

single phoneme

(right)

Method - Preprocessing

분석 구현 흐름 · conceptual outline

raw EEG → event alignment → resample & filter → epoching → bad-epoch rejection → ICA/ICLabel → ML-ready tensor

1

Signal 표준화

- Event 확인 및 자극 시점 정렬
- Resampling: 2,000 → 256 Hz
- Notch filtering: 59–61 Hz
- Band-pass filtering: 0.1–100 Hz

2

Epoching & Quality Control

- stim_1 기준 Epoching
- 1 epoch = 1 s, 256 samples
- Robust Signal-Quality Metrics

3

Artifact Cleaning & Conversion

- ICA로 독립 성분 분리
- ICLabel로 artifact 성분 판별
- 비신경성 성분 제거
- Trials × 61 Channels × 256 Samples

Method - Data Quality Validation

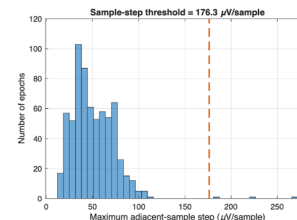
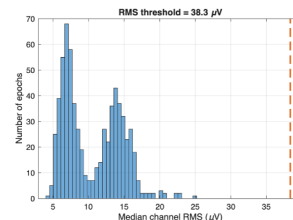
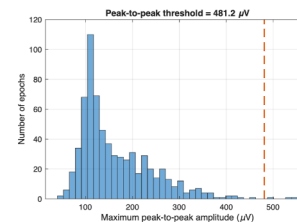
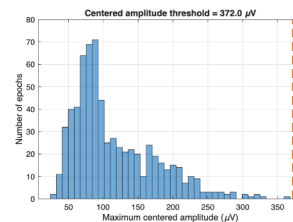
1. Epoch-level QC

- Use : Robust signal-quality metrics
- Threshold: Median + 6 × Robust SD

S02_5_bad_epoch_thresholds

Metric	AutomaticThreshold	Median	RobustStandardDeviation	Unit
MaxCenteredAbs	371.982719335079	94.389479637146	46.2655399496555	μV
MaxPeakToPeak	481.181607009888	138.202651977539	57.1631591720581	μV
MedianChannelRMS	38.3387565623482	9.43671929871714	4.81700621060517	μV
MaxSampleStep	176.307672012901	46.0797710418701	21.7046501618385	μV/sample

[Automatic Bad-Epoch Rejection Thresholds]



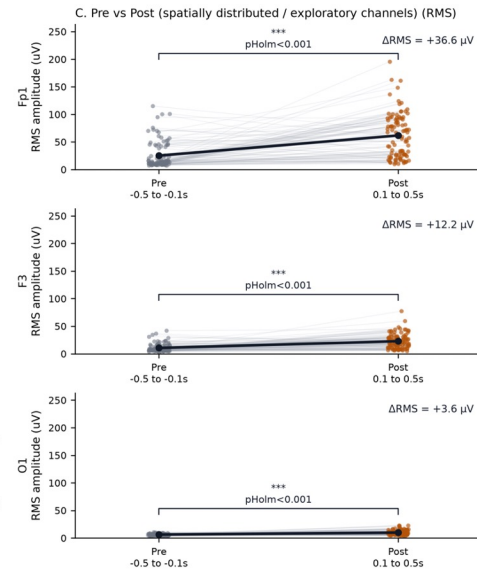
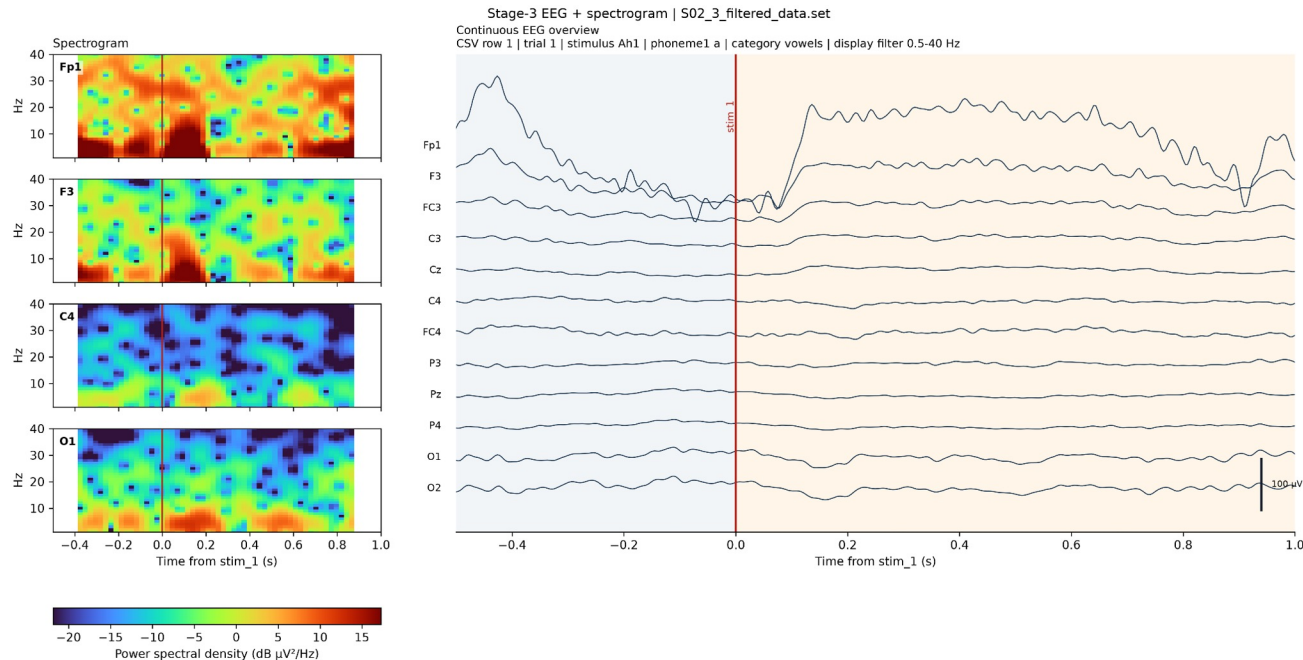
2. ICA-level QC

- Use : ICLabel-based classification

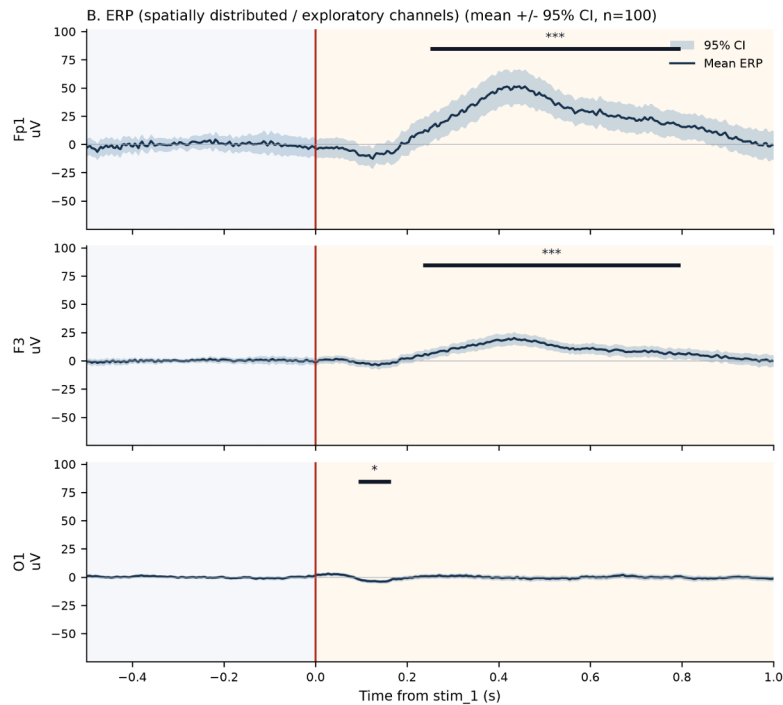
S02_7_ica_component_report

Component	MostLikelyClass	MostLikelyProbability	BrainProbability	MuscleProbability	EyeProbability	HeartProbability	LineNoiseProbability	ChannelNoiseProbability	OtherProbability	CheckedAmongFirst35	AutomaticallyRemoved
1	Eye	0.953694	0.003151193	0.02544772	0.953694	0.0003213965	0.0002083843	0.00214142	0.01503593	1	1
2	Eye	0.9848065	0.004639731	0.002100364	0.9848065	0.0001408733	5.459438E-05	0.0004825056	0.007775463	1	1
3	Other	0.9577852	0.01784888	0.003905366	0.0003473679	0.008988831	0.006593634	0.004530751	0.9577852	1	0
4	Other	0.9329869	0.03855471	0.006734762	0.002155683	0.004568655	0.005215362	0.009784021	0.9329869	1	0

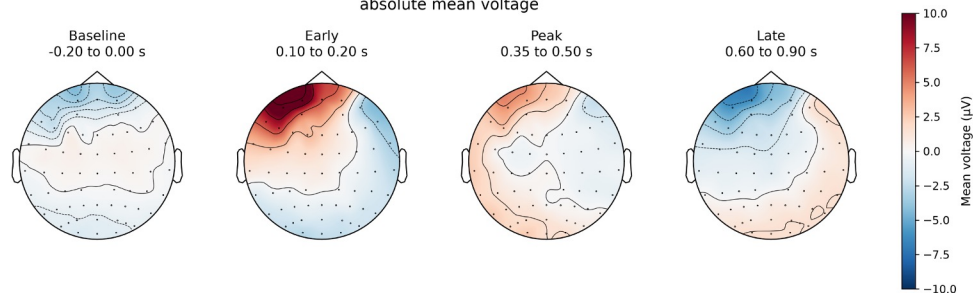
Results - Stimulus-Locked Characterization



Results - Stimulus-Locked Characterization



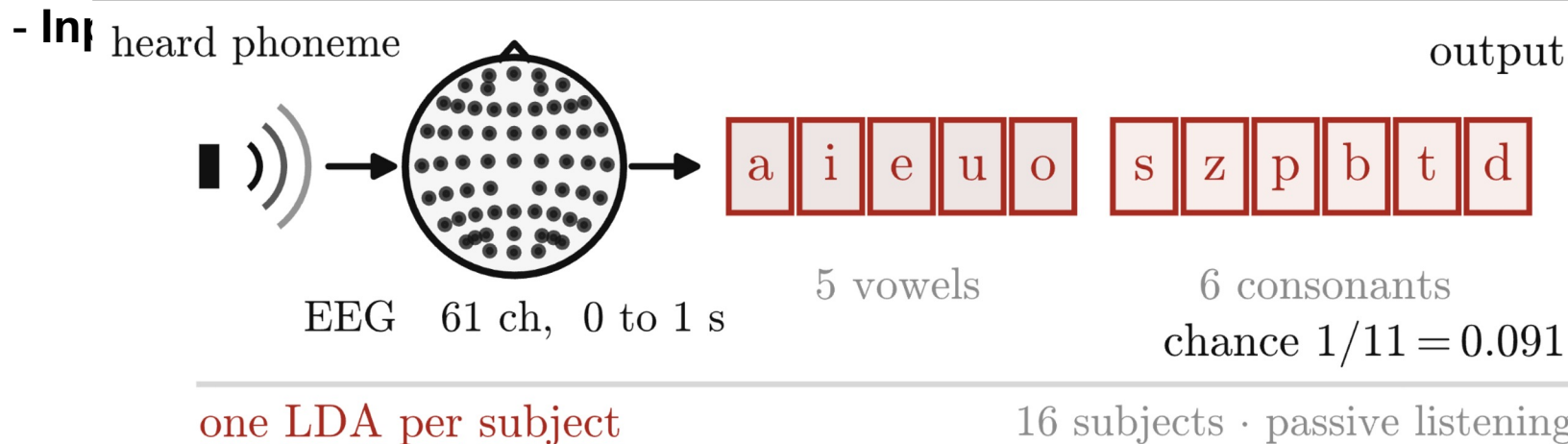
ERP topographic snapshots | task = phonemes | tms0 + control | n = 80
absolute mean voltage



ML Machine Learning-Based Model

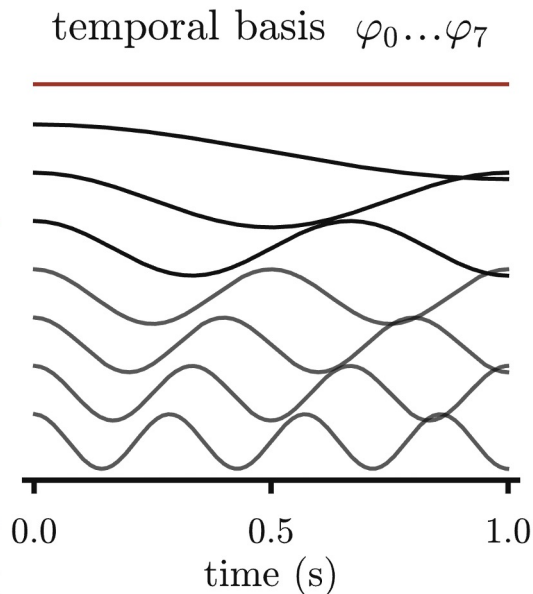
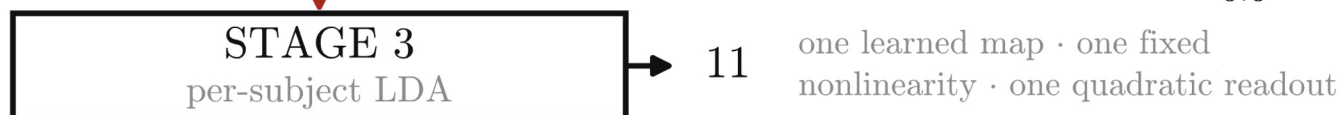
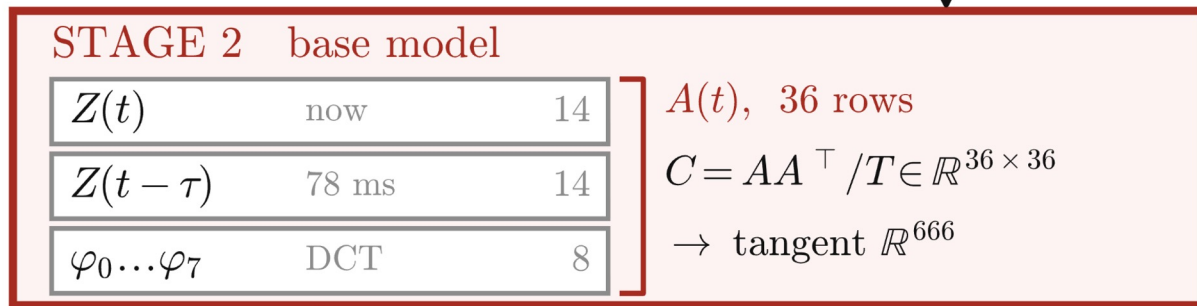
- Goal of our model:

Classify 11 phonemes (a, i, e, u, o, s, z, p, b, t, and d) — chance level:
0.091



ML Machine Learning-Based Model

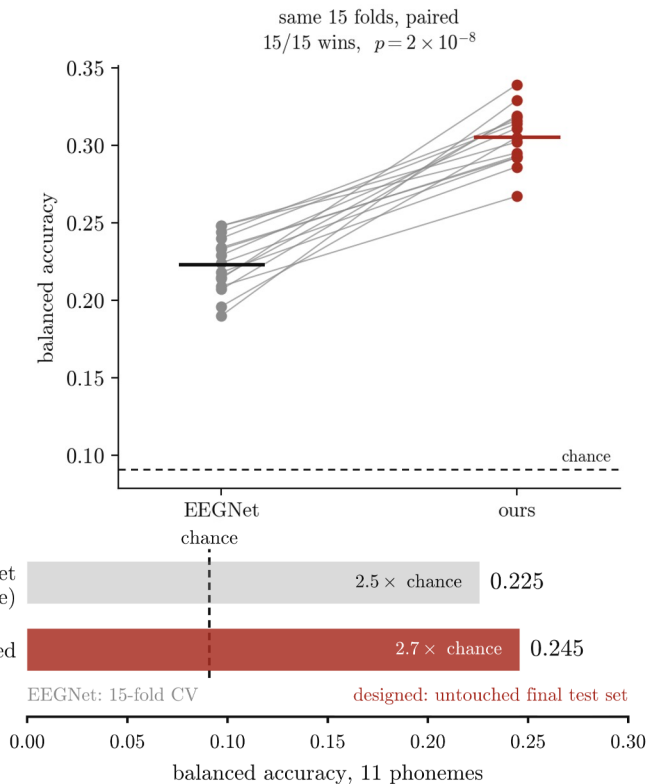
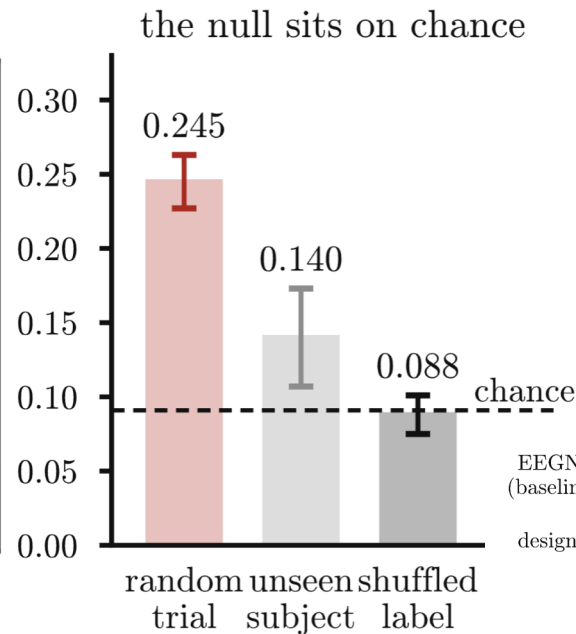
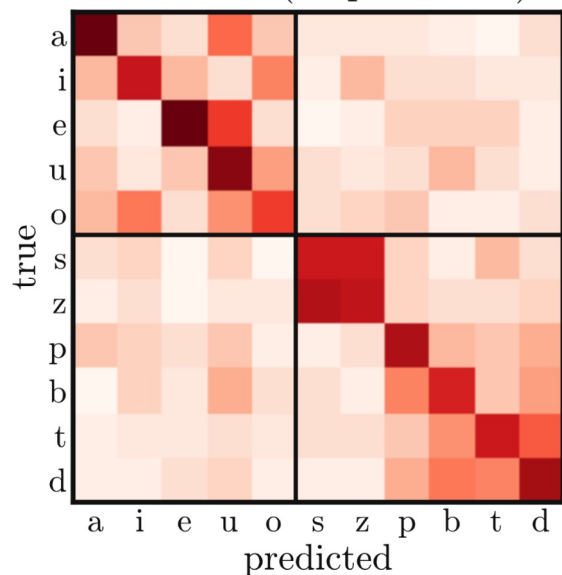
- Model Structure:



ML Machine Learning-Based Model

- Result:

confusion (11 phonemes)



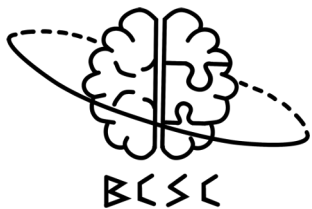
Limitation

Task-related Overlap

- Task 중 single phenomenon에서 참가자는 단일 음소를 들은 직후 그대로 발음함.
- speech production-related activity가 comprehension related potential과 시간적으로 겹칠 수 있음.

Interpretation limit

- EEG response에는 청각·언어·운동 처리가 함께 반영될 수 있음.
- ERP 변화와 분류 성능을 순수한 내적 발화 신호로 해석하기 어려움.



Thank You

Brain and Cognitive Science Community 10th

Team leader: 문경은 | Team members: 구민주 김소영 노현이 윤시원 이재용 조연우 채보경